

Deciphering the molecular grammar of gap junctions using integrative structural bioinformatics

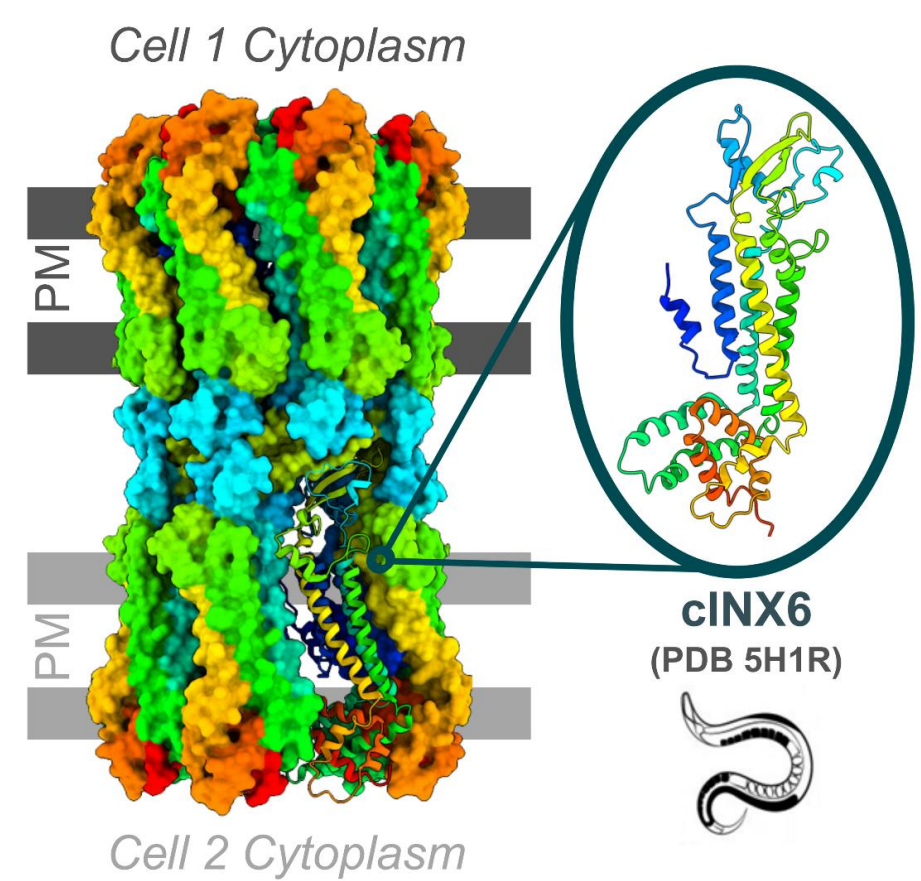
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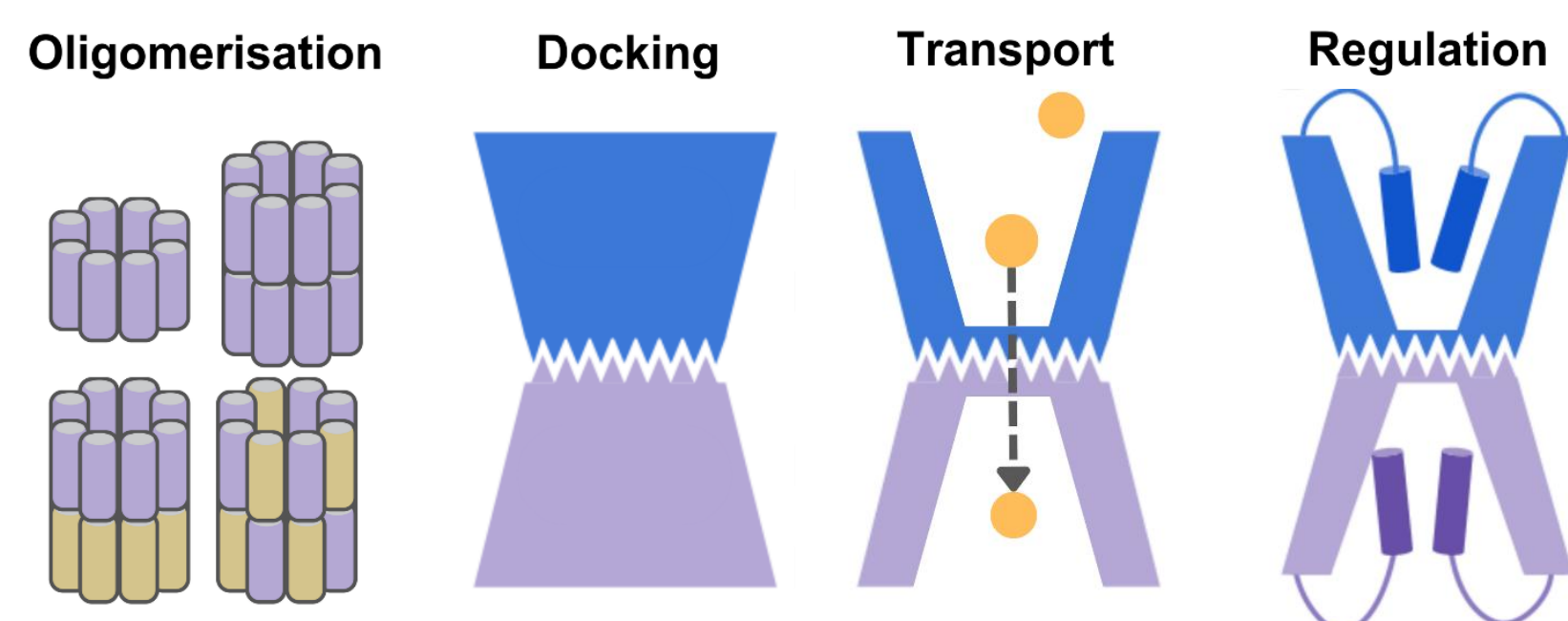
Introduction

Gap junctions are **intercellular channels** facilitating a direct connection between 2 neighbouring cells

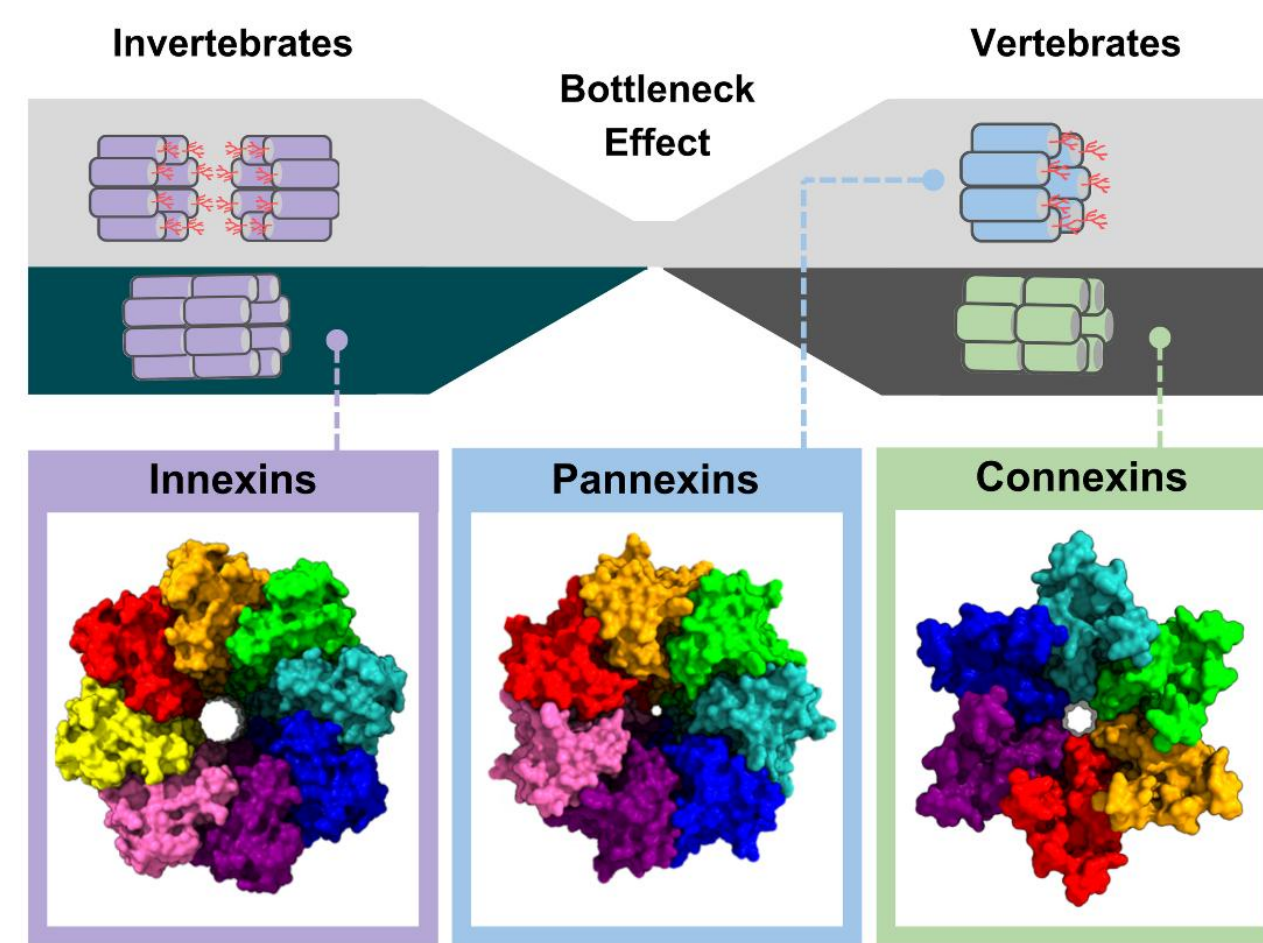


Why are gap junctions (structurally) interesting ?

Gap junctions have **several properties** encoded into a single **monomer sequence**

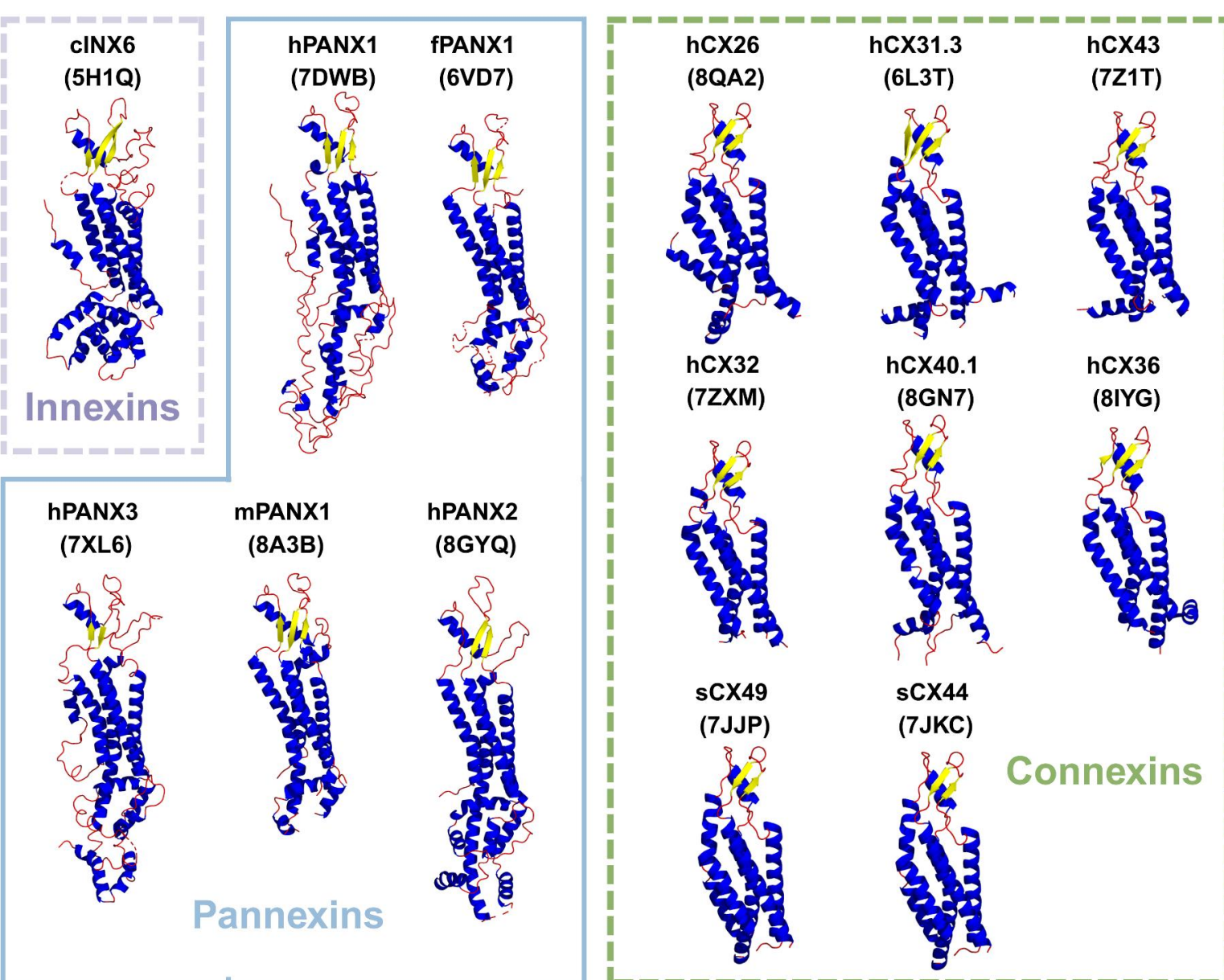


Gap Junction Evolution (Welzel et al., 2022)

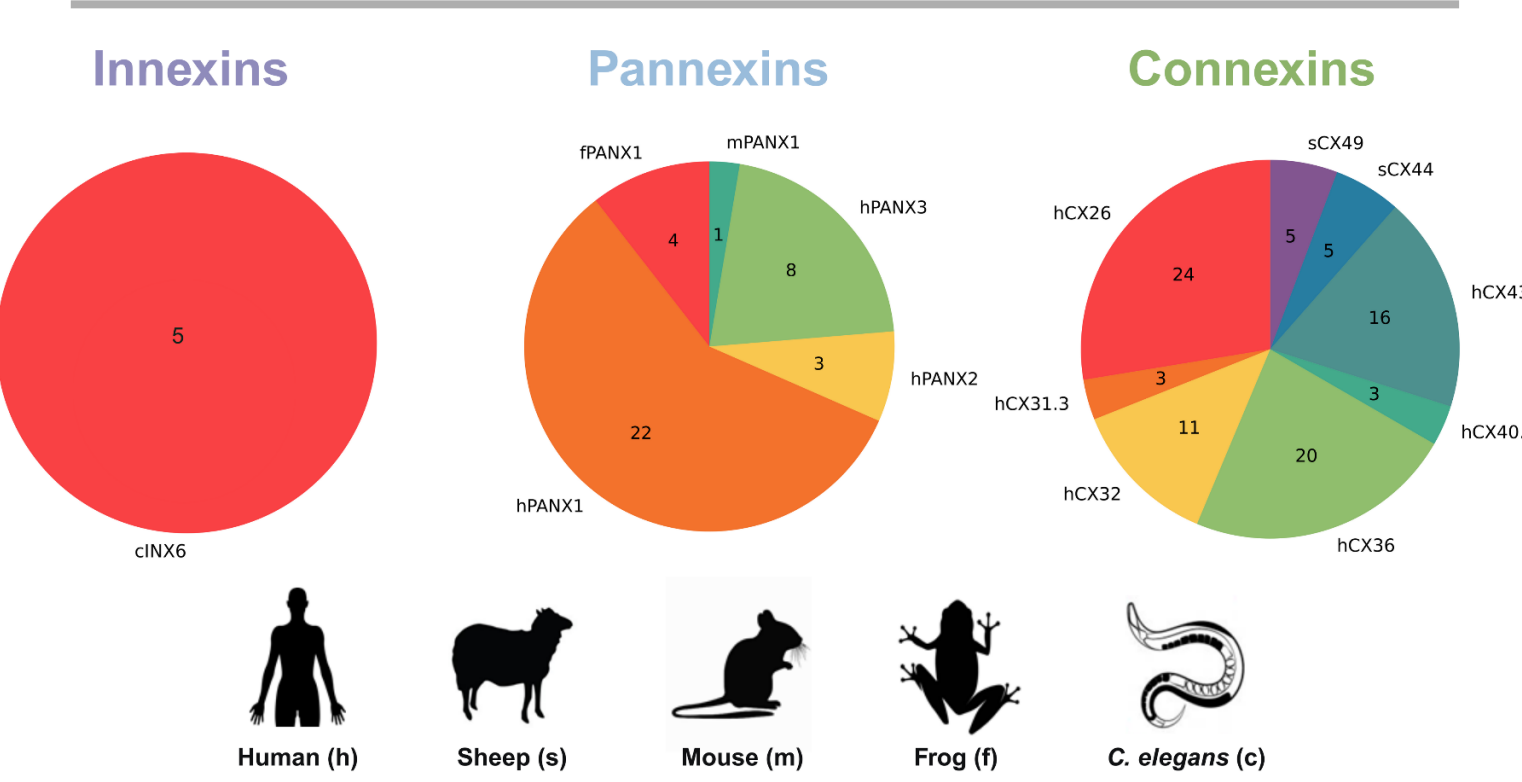


Dataset

Representative monomer structure for each protein in the study



Number of PDB structures for different gap junction proteins



Objectives

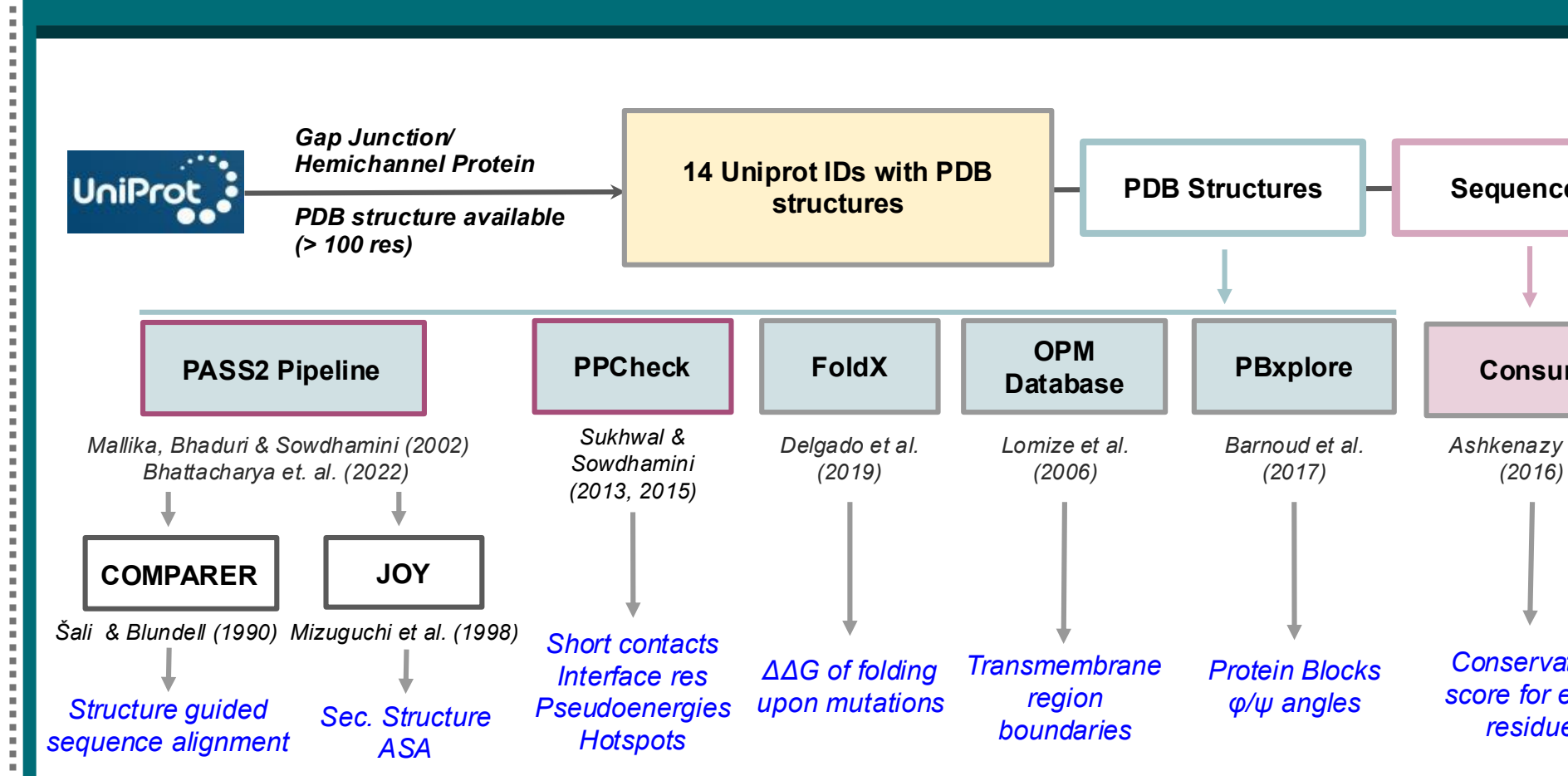
- Comparing structural properties across the three gap junction protein families
- Identifying the set of residue positions which potentially contribute to a junctional property
- Create a **sequence-structure-function map** which can be used -

As a blueprint for designing new junctions

To identify mutations to selectively perturb junctional properties

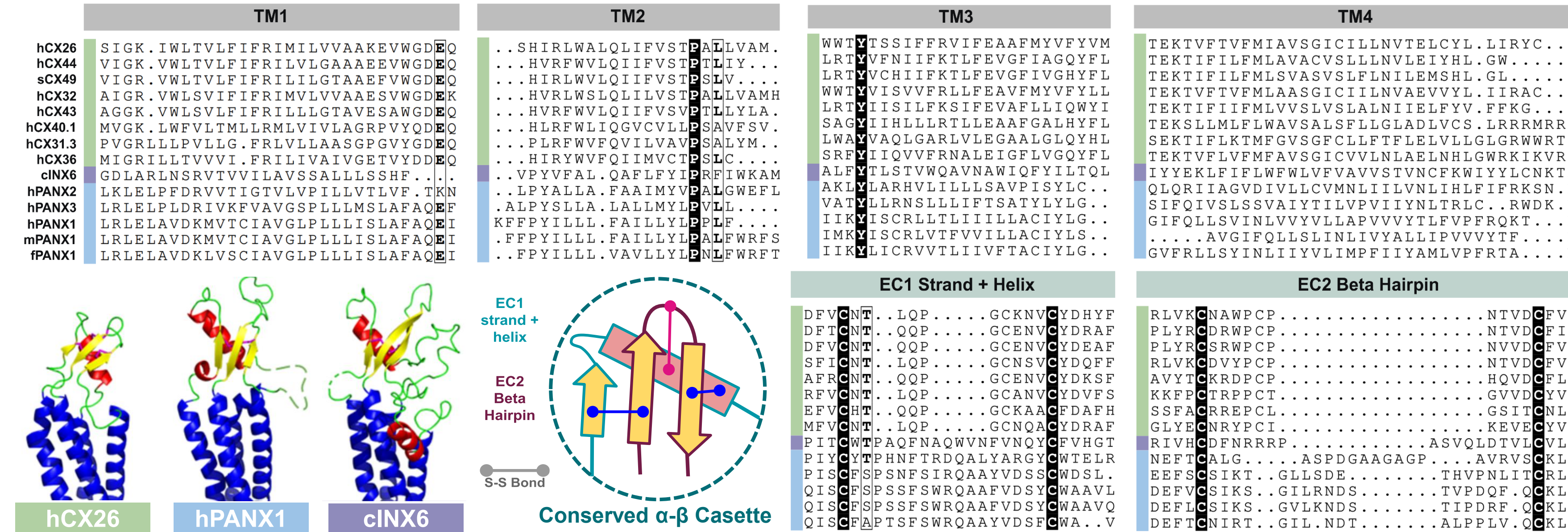


Methodology

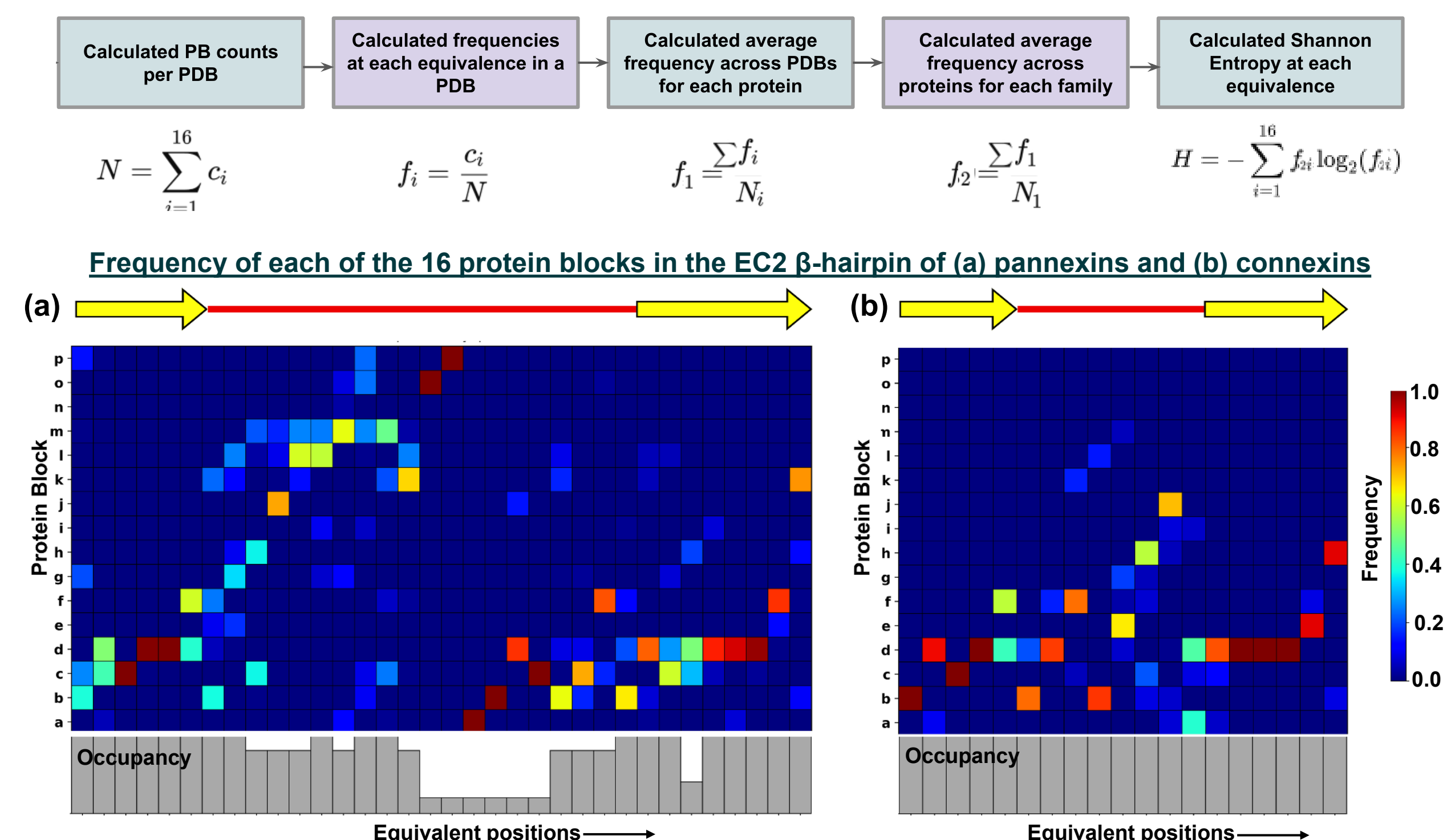


Results

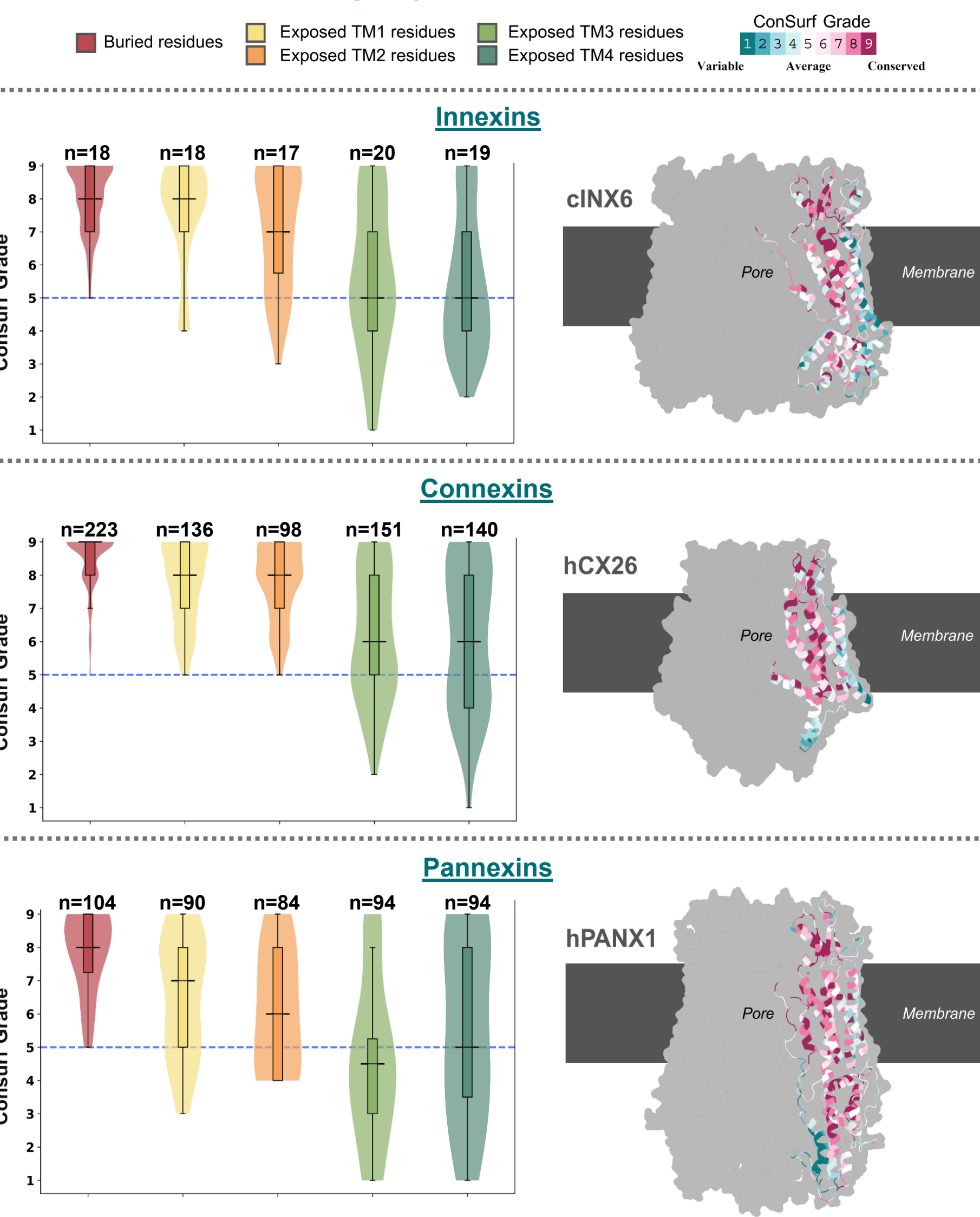
Structure-based sequence alignments show multiple equivalences across gap junction protein families



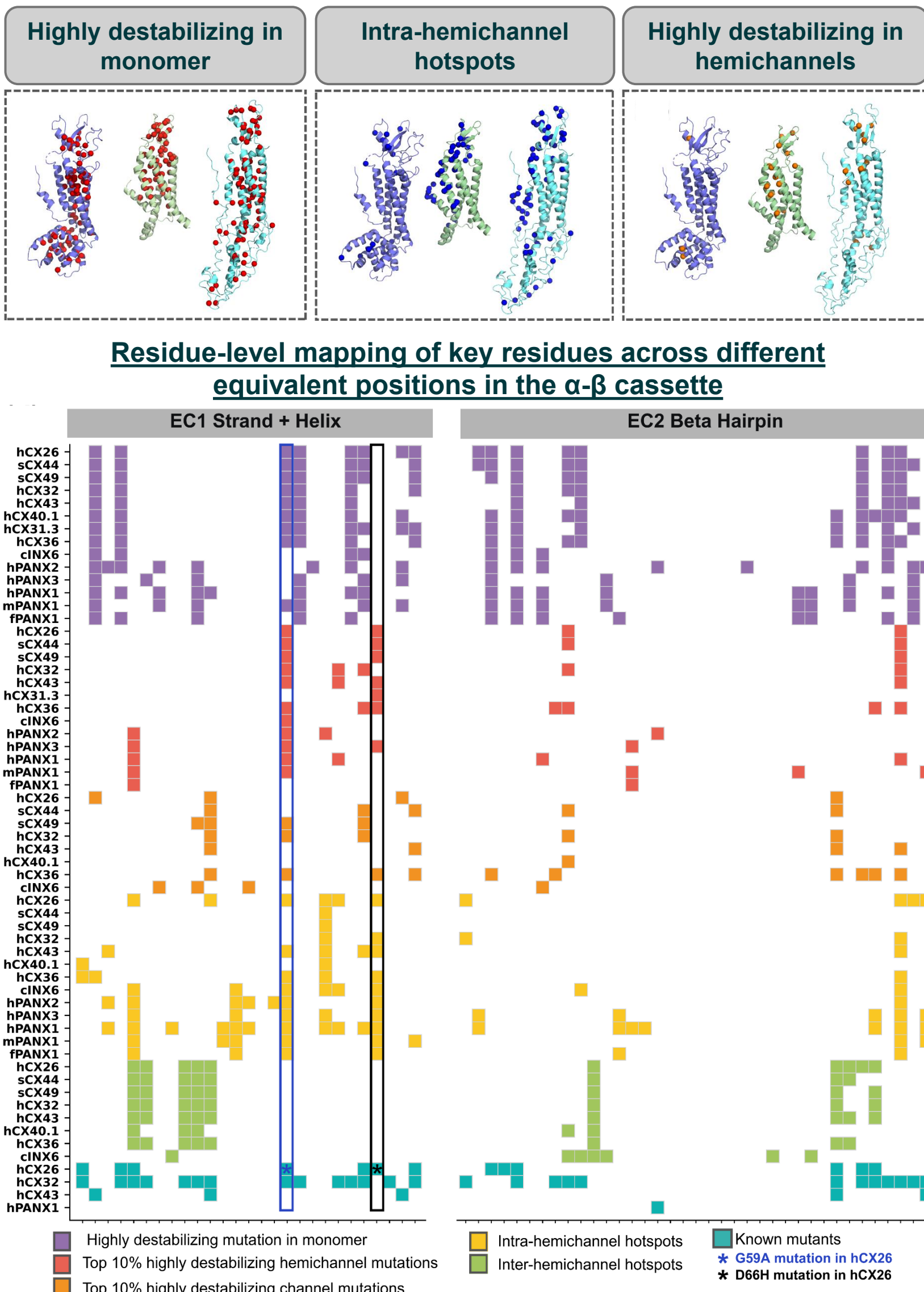
Protein-block patterns shed light on the local structural variability of the extracellular α-β cassette



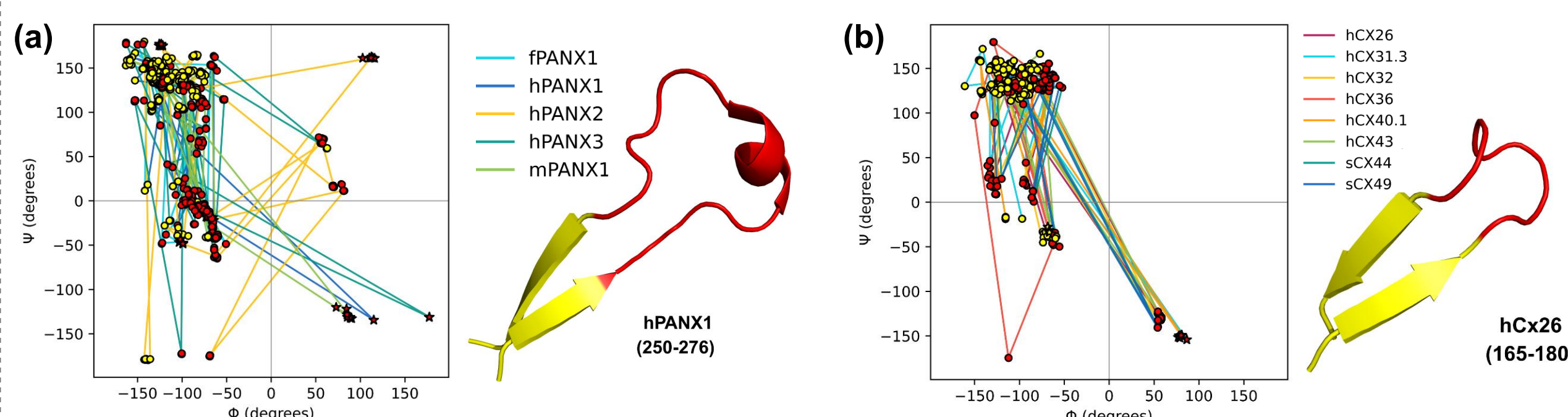
Residue conservation patterns across transmembrane helices in gap-junction protein families



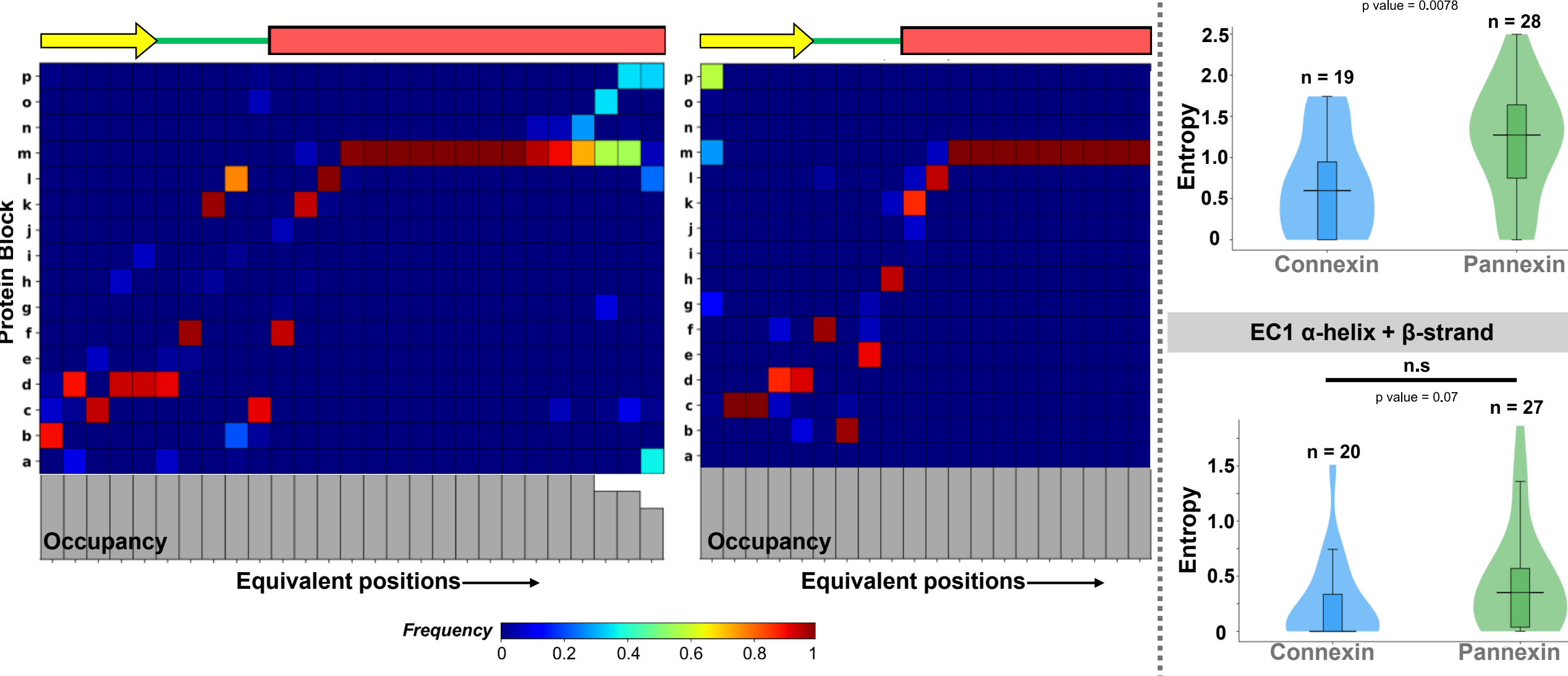
Key residues driving oligomerization map to equivalent positions



Ramachandran plots for (a) pannexin and (b) connexin β-hairpin residues



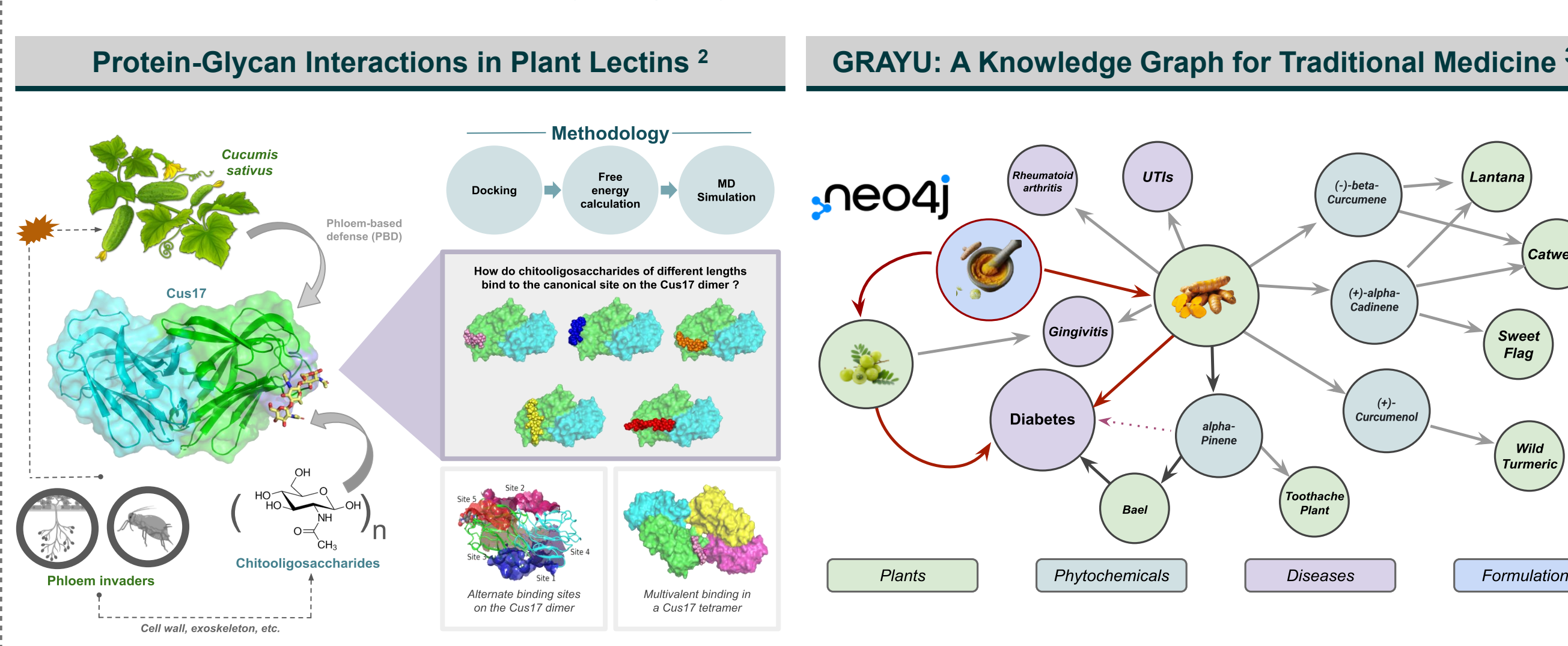
Frequency of each of the protein blocks in the EC1 α-helix + β-strand of (left) pannexins and (right) connexins



Conclusions & Perspectives

- Despite negligible sequence similarity, we established a **structural consensus across divergent families**
- High conservation in pore-lining TM1/TM2 reflects **selectivity constraints**, whereas the conformational flexibility of pannexin EC2 loops suggests a relaxation of structural rigidity following their **evolutionary divergence from docking functions**
- Our pipeline successfully **identified universal energetic hotspots** that coincide with known disease-causing mutations, validating its ability to pinpoint residues critical for channel assembly
- This "molecular grammar" provides a roadmap for the **in-silico design of synthetic intercellular conduits** and the development of **precision modulators** for gap junction-related pathologies

Interested in more than just gap junctions? Here is some of our other work:



References

- Pathak A and Sowdhamini R. Computational structural analysis sheds light on the molecular grammar of gap junction proteins. *Biochimie* (2025)
- Tiwari V*, Pathak A*, Poojary T, Sowdhamini R, Surolia A. Exploration of structural space of Cus17 lectin with glycans using molecular docking and simulation studies. *Computational and Structural Biotechnology Journal* (2025)
- Joshi S*, Pathak A*, Regati DR*, Menon R, Ajith DS, Sheshadri A, Viswanathan N, Ray P, Koul V, Panda P, Bhambore SA, Verma S, Sinha A, Shafi KM, Pavalam M, Sowdhamini R. GRAYU: Graph-based Database Integrating Ayurvedic formulations, medicinal plants, phytochemicals and diseases. *Frontiers in Pharmacology* (2026)

Acknowledgements

